Data-Backed Journal-Style Replication of *Something Old, Something New*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 2,174 observations across 162 entities, and the batch headline estimate is 0.0052 for search_scope with p = 0.6153. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: search depth; search scope; new product introduction; firm-year innovation; local analogue

1 Introduction

This replication begins from a paper-first reading of Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether search depth and search scope are distinct dimensions and how

they shape subsequent innovation. The study is worth revisiting because The replication matters because the original paper's distinction between search depth and search scope is widely cited, yet the empirical mechanism depends heavily on how those constructs are operationalized. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 2,174 observations across 162 entities and spans 1986 to 2015 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. The original paper distinguishes search depth from search scope. The study is longitudinal and firm-year based. Innovation quantity and quality are both part of the argument. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local

replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Depth and scope are distinct dimensions. The local reconstruction therefore adopts the following bounded decision: The local panel maps depth and scope to executable recombination constructs. This bridge is treated as a visible design choice, not as a hidden substitution. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The original design is firm-year based. The local reconstruction therefore adopts the following bounded decision: The firm-year structure is preserved exactly. This bridge is treated as a visible design choice, not as a hidden substitution. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Innovation outcomes include quantity and quality. The local reconstruction therefore adopts the following bounded decision: Patent counts and patent-impact totals substitute for product introductions. This bridge is treated as a visible design choice, not as a hidden substitution.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Firm-year panel aggregated from the local patent-level recombination dataset. Search depth measured from prior organizational recombination experience. Search scope measured from breadth of recombination in the current year. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv; Coefficient export is backed by patent literature/daily_replications/2026-04-04/katila-ahuja-

2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv; Sample summary is backed by patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv. The resulting executable panel supplies 2,174 observations across 162 entities and covers 1986 to 2015 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. The local sample frame is Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions. The observed time window in the executable panel is 1986 to 2015 in the local executable panel. The outcome is Next-year patent quantity and next-year total patent impact. The main executable panel is stored at patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv and the principal coefficient export used in the paper is stored at patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv . The local analysis file contains 2,174 observations drawn from 162 entities. The observed time span runs from 1,986 to 2,015. Selected sample moments are search_depth = 0.6637, search_scope = 4.2347, next_patents = 10.6026 . These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed.

3.1 Variable Construction and Derivation

The construct Search depth is operationalized locally as Mean prior organizational experience and appears in the executable dataset as search_depth. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Average the current-year patent-level

organizational experience within GVKEY-year.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Search scope is operationalized locally as Mean breadth and appears in the executable dataset as search_scope. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Average current-year breadth within GVKEY-year.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Current innovation scale is operationalized locally as Current-year patent count and appears in the executable dataset as current_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count current-year patents.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Next-period quantity is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Shift the patent count one year forward within GVKEY.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Next-period impact is operationalized locally as Next-year total impact and appears in the executable dataset as next_total_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Shift the impact total one year forward within GVKEY.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation.

4 Empirical Strategy and Identification

The estimand targeted in this manuscript is Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that

embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions., and the empirical window is 1986 to 2015 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Next-year patent quantity and next-year total patent impact., and the main explanatory design is Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Current innovation scale and field-experience controls with firm and year effects.. The crucial exclusion or substitution rule is: The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

$$E[\text{NextPatents}_{it} \mid X_{it}] = \exp\left(\beta_1 \text{Depth}_{it} + \beta_2 \text{Depth}_{it}^2 + \beta_3 \text{Scope}_{it} + \beta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t\right), \tag{1}$$

$$\log(1 + \text{NextImpact}_{it}) = \theta_1 \text{Depth}_{it} + \theta_2 \text{Depth}_{it}^2 + \theta_3 \text{Scope}_{it} + \theta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{2}$$

5 Results

5.1 Main Estimates

The central local estimate is 0.0052 on search_scope (p = 0.6153). The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. Log(1 + current patents) is estimated at 0.6357 with p = ¡0.001. Depth x scope is estimated at -0.0216 with p = 0.0982. Search scope is estimated at 0.0184 with p = 0.1671. Table 1 reports the compact coefficient pattern most relevant to the source paper's central claim. The broader coefficient dump, descriptive statistics, grouped summaries, and audit material are retained in the appendix so that the main text can focus on interpretation.

model	Term	estimate	std_error	p_value
katila_ahuja_next_impact	Search depth	-0.1070	0.1214	0.3854
katila_ahuja_next_impact	Search depth squared	0.0388	0.0322	0.2375
katila_ahuja_next_impact	Search scope	0.0184	0.0130	0.1671
katila_ahuja_next_impact	Log(1 + current patents)	0.6357	0.0617	0.0000
katila_ahuja_next_impact	Mean field experience	0.0271	0.0350	0.4444
katila_ahuja_next_impact	Depth x scope	-0.0216	0.0126	0.0982

Key coefficient table used in the main text.

5.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log(1 + current patents) is estimated at 0.6357 with a p-value of ¡0.001; Depth x scope is estimated at -0.0216 with a p-value of 0.0982; Search scope is estimated at 0.0184 with a p-value of 0.1671; Search depth squared is estimated at 0.0388 with a p-value of 0.2375 . Within the focal specification, the sharpest local signal comes from Log(1 + current patents), which is estimated at 0.6357 with p = ¡0.001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim

rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 10.6026 and standard deviation 21.2666 in the local panel, while the main regressor Search depth has mean 0.6637 and standard deviation 0.6622. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 5.3149 to 17.4105 while the mean regressor ranges from 0.1128 to 1.6987. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while search scope is comparatively muted in the executable patent-based panel. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

7 Discussion

The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while

search scope is comparatively muted in the executable patent-based panel. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local analogue uses patent outcomes rather than product-introduction outcomes. The second limitation concerns sample translation. Search variables are mapped from recombination measures already in the workspace. The third limitation concerns interpretation. The exercise is strongest on conceptual mechanism and panel structure, not exact outcome identity.

8 Conclusion

The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Reading Protocol

This appendix records the paper-first workflow followed before any manuscript text was written. The sequence starts from the source article, then maps required empirical objects to verified local files, and only then proceeds to estimation and interpretation.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/KatilaAhujaamj02.txt
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

B Appendix B: Data Lineage and File Manifest

This appendix documents the data lineage as a sequence rather than as a pile of filenames. The source article anchors the theory and published design. The executable local files then provide the panel, coefficient export, sample reconstruction log, and, where available, fit statistics used in the manuscript. Source article: patent literature/KatilaAhujaamj02.txt. Its role in the manuscript is Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices. Analysis panel: patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv.
Its role in the manuscript is Estimate the local model and regenerate descriptive statistics. Coefficient export: patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv. Its role in the manuscript is Recover the reported local coefficients and p-values. Sample summary: patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv. Its role in the manuscript is Audit sample construction and raw grouped moments. The purpose of laying out this chain in prose is to show that the manuscript's claims travel through a verified sequence of evidence. In other words, the paper does not jump directly from the published theory to the final coefficient table. It moves through the exact local files needed to rebuild the analysis for Katila and Ahuja (2002).

File	Role
katila_ahuja_2002_analysis_dataset.csv	Firm-year search-behavior panel
katila_ahuja_2002_model_coefficients.csv	Quantity and impact coefficient table
katila_ahuja_2002_sample_summary.csv	Sample statistics

Compact list of core local files used for this manuscript.

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/KatilaAhujaamj02.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

C Appendix C: Replication Boundary and Decision Register

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions. over 1986 to 2015 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Next-year patent quantity and next-year total patent impact. and the main explanatory design is Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Current innovation scale and field-experience controls with firm and year effects. The main exclusion or substitution rule is The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

DirectSourceFact	LocalReplicationDecision
Depth and scope are distinct dimensions.	The local panel maps depth and scope to executable recombination constructs.
The original design is firm-year based.	The firm-year structure is preserved exactly.
Innovation outcomes include quantity and quality.	Patent counts and patent-impact totals substitute for product introductions.

Direct source facts versus explicit local replication decisions.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.
Unit of analysis	Firm-year
Sample frame	Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions.
Time window	1986 to 2015 in the local executable panel.
Outcome definition	Next-year patent quantity and next-year total patent impact.
Main explanatory design	Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.
Controls and fixed effects	Current innovation scale and field-experience controls with firm and year effects.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value
rows	2174.0000
unique_firms	162.0000
year_min	1986.0000
year_max	2015.0000
mean_search_depth	0.6637
mean_search_scope	4.2347
mean_next_patents	10.6026

Detailed sample reconstruction table.

E Appendix E: Variable Mapping and Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Katila and Ahuja (2002). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Search depth, the manuscript uses Mean prior organizational experience under the exact local implementation search_depth. The derivation step is: Average the current-year patent-level organizational experience within GVKEY-year.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Search scope, the manuscript uses Mean breadth under the exact local implementation search_scope. The derivation step is: Average current-year breadth within GVKEY-year.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Current innovation scale, the manuscript uses Current-year patent count under the exact local implementation current_patents. The derivation step is: Count current-year patents.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Next-period quantity, the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Shift the patent count one year forward within GVKEY.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Next-period impact, the manuscript uses Next-year total impact under the exact local implementation next_total_impact. The derivation step is: Shift the impact total one year forward within GVKEY.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search depth	Mean prior organizational experience	search_depth	Average the current-year patent-level organizational experience within GVKEY-year.
Search scope	Mean breadth	search_scope	Average current-year breadth within GVKEY-year.
Current innovation scale	Current-year patent count	current_patents	Count current-year patents.
Next-period quantity	Next-year patent count	next_patents	Shift the patent count one year forward within GVKEY.
Next-period impact	Next-year total impact	next_total_impact	Shift the impact total one year forward within GVKEY.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Search depth	2174.0000	0.6637	0.6622	0.1386	0.5493	0.9635	0.0000	4.76
Search scope	2174.0000	4.2347	2.8896	2.8182	3.4264	4.6667	2.0000	52.00
Current patents	2174.0000	10.5437	20.1122	2.0000	5.0000	11.0000	1.0000	399.00
Current total impact	2174.0000	56.5745	142.3809	5.0000	17.0000	47.0000	0.0000	3058.00
Mean field experience	2174.0000	1.4705	1.1482	0.6931	1.3041	1.9634	0.0000	7.55
Next-year patents	2174.0000	10.6026	21.2666	2.0000	4.0000	11.0000	1.0000	399.00
Next-year total impact	2174.0000	56.6297	144.9651	5.0000	16.0000	46.0000	0.0000	3058.00

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Search depth	Search scope	Current patents	Current total impact	Mean field experience	Next-year patents	Next-year total impact
Search depth	1.000	-0.011	0.184	0.138	0.735	0.158	0.106
Search scope	-0.011	1.000	0.010	0.146	-0.242	0.004	0.092
Current patents	0.184	0.010	1.000	0.612	0.062	0.746	0.513
Current total impact	0.138	0.146	0.612	1.000	0.042	0.482	0.670
Mean field experience	0.735	-0.242	0.062	0.042	1.000	0.046	0.020
Next-year patents	0.158	0.004	0.746	0.482	0.046	1.000	0.630
Next-year total impact	0.106	0.092	0.513	0.670	0.020	0.630	1.000

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4	V5
1986.0000	100.0000	0.2522	2.7740	7.7000	33.2900
1987.0000	103.0000	0.3220	2.9110	10.2136	45.3592
1988.0000	105.0000	0.3484	3.0106	13.1714	51.6571
1989.0000	98.0000	0.4390	3.1399	13.5102	61.0510
1990.0000	111.0000	0.4669	3.0152	11.9640	56.5225
1991.0000	109.0000	0.5001	3.1953	12.2110	61.1927
1992.0000	98.0000	0.4820	3.3594	13.5408	77.0510
1993.0000	100.0000	0.5504	3.4467	13.2200	82.5600
1994.0000	104.0000	0.5512	3.6508	11.6442	72.5481
1995.0000	98.0000	0.5892	3.7678	6.9490	45.8367
1996.0000	83.0000	0.6681	4.1041	7.6265	45.4217
1997.0000	80.0000	0.8318	4.0021	4.8500	30.2750
1998.0000	65.0000	0.6553	3.9114	9.2000	50.7077
1999.0000	78.0000	0.6504	3.7812	8.1410	45.8590
2000.0000	73.0000	0.8168	3.9949	8.3014	42.4384
2001.0000	72.0000	0.7542	4.2585	8.8056	42.2083
2002.0000	65.0000	0.8268	4.4429	8.0462	36.2308
2003.0000	62.0000	0.8369	4.8452	5.2581	19.5000
2004.0000	59.0000	0.7436	4.8832	5.0847	23.0678
2005.0000	48.0000	0.7782	5.1188	6.8125	33.2083
2006.0000	51.0000	0.8915	5.2182	4.9216	17.0392
2007.0000	43.0000	0.8791	6.2118	12.6047	48.6047
2008.0000	38.0000	0.9390	5.6773	5.2895	25.4737
2009.0000	43.0000	1.0399	6.1656	11.3488	49.2093

Period	V1	V2	V3	V4	V5
2010.0000	50.0000	0.9987	6.7648	12.9600	83.4800
2011.0000	51.0000	0.9412	7.1105	20.7059	83.2157
2012.0000	52.0000	1.0293	6.9358	16.5192	88.8462
2013.0000	50.0000	1.0361	6.7054	14.3400	180.3600
2014.0000	50.0000	1.1223	6.3393	21.3600	141.9400
2015.0000	35.0000	1.2033	6.8650	14.8857	57.1429

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	870.0000	0.1128	5.3149
2.0000	506.0000	0.5658	11.7273
3.0000	363.0000	0.8803	17.4105
4.0000	435.0000	1.6987	14.1885

Quantile summary for the main regressor.

J Appendix J: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
search_depth	-0.2243	0.1001	-2.2411	0.0250	katila_ahuja_next_pate
I(I(search_depth^2))	0.1059	0.0299	3.5451	0.0004	katila_ahuja_next_pate
search_scope	0.0052	0.0104	0.5025	0.6153	katila_ahuja_next_pate
log1p(current_patents)	0.6654	0.0652	10.2026	0.0000	katila_ahuja_next_pate
mean_field_exp	-0.0429	0.0262	-1.6360	0.1018	katila_ahuja_next_pate
search_depth:search_scope	-0.0115	0.0103	-1.1143	0.2651	katila_ahuja_next_pate
search_depth	-0.1070	0.1214	-0.8812	0.3854	katila_ahuja_next_impa
I(I(search_depth^2))	0.0388	0.0322	1.2061	0.2375	katila_ahuja_next_impa
search_scope	0.0184	0.0130	1.4170	0.1671	katila_ahuja_next_impa
log1p(current_patents)	0.6357	0.0617	10.3089	0.0000	katila_ahuja_next_impa
mean_field_exp	0.0271	0.0350	0.7753	0.4444	katila_ahuja_next_impa
search_depth:search_scope	-0.0216	0.0126	-1.7086	0.0982	katila_ahuja_next_impa

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Search depth	0.0000	0.0000
Search scope	0.0000	0.0000
Current patents	0.0000	0.0000
Current total impact	0.0000	0.0000
Mean field experience	0.0000	0.0000
Next-year patents	0.0000	0.0000
Next-year total impact	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
10499.0000	30.0000	1986.0000	2015.0000
10983.0000	30.0000	1986.0000	2015.0000
1300.0000	30.0000	1986.0000	2015.0000
2285.0000	30.0000	1986.0000	2015.0000
2403.0000	30.0000	1986.0000	2015.0000
3532.0000	30.0000	1986.0000	2015.0000
4087.0000	30.0000	1986.0000	2015.0000
4321.0000	30.0000	1986.0000	2015.0000
4503.0000	30.0000	1986.0000	2015.0000
4839.0000	30.0000	1986.0000	2015.0000
5047.0000	30.0000	1986.0000	2015.0000
5439.0000	30.0000	1986.0000	2015.0000
6066.0000	30.0000	1986.0000	2015.0000
6266.0000	30.0000	1986.0000	2015.0000
7435.0000	30.0000	1986.0000	2015.0000
8762.0000	30.0000	1986.0000	2015.0000
2817.0000	29.0000	1986.0000	2015.0000
3835.0000	29.0000	1986.0000	2015.0000
6435.0000	29.0000	1986.0000	2015.0000
7257.0000	29.0000	1986.0000	2015.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.	Direct source fact	patent literature/KatilaAhujaamj02.txt	Recovered from the source article before local writing begins.
The local executable panel contains 2,174 observations across 162 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.cs — patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_sample_summary.c	Backed by the verified analysis panel and sample audit.
The local headline estimate is 0.0052 for search_scope with p = 0.6153.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_model_coefficients	Backed by the exported coefficient file used in the paper.

Claim	EvidenceType	EvidenceFile	AuditNote
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

O Appendix O: Manuscript Metadata

Field	Value
Paper	Katila and Ahuja (2002)
Source title	Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction
Authors	Riitta Katila and Gautam Ahuja
Outlet	Academy of Management Journal
Batch date	2026-04-04
Rows	2,174
Entities	162
Headline term	search_scope
Headline estimate	0.0052
Headline p-value	0.6153
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-ahuja-2002-search-behavior/results/katila_ahuja_2002_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

P Appendix P: Replication Summary Note

Research question: The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation. Local design: The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. Estimand: Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. Local result focus: The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. Comparison note: The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while search scope is comparatively muted in the executable patent-based panel.

References

[1] Riitta Katila and Gautam Ahuja. Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194, 2002.